



Week 8.4

Recap Everything, Build PayTM Frontend

In this lecture, Harkirat guides us through an **end-to-end tutorial** on building a comprehensive **full-stack application** resembling **Paytm**. While there are no specific notes provided for this section, a mini guide is outlined below to assist you in navigating through each step of the tutorial. Therefore, it is strongly advised to actively follow along during the lecture for a hands-on learning experience.

It's important to note that this session primarily focuses on the frontend section of the application. For the backend portion, be sure to check out the content covered in the previous 8.2 lecture.

Recap Everything, Build PayTM Frontend

Step 1 - Add routing to the react app

Solution

Step 2 - Create and hook up Signup page

Step 3 - Create the signin page

Step 4 - Dashboard page

Step 5 - Auth Components

1. Heading component

Code

2. Sub Heading component

Code

3. InputBox component

Code

4. Button Component

Code

5. BottomWarning

Code

Full Signup component

Code

Full Signin component

Code

Step 6 - Signin-ed Comonents

1. Appbar

Code

2. Balance

Code

3. Users component

Code

4. SendMoney Component

Code

Step 7 - Wiring up the backend calls

Step 1 - Add routing to the react app

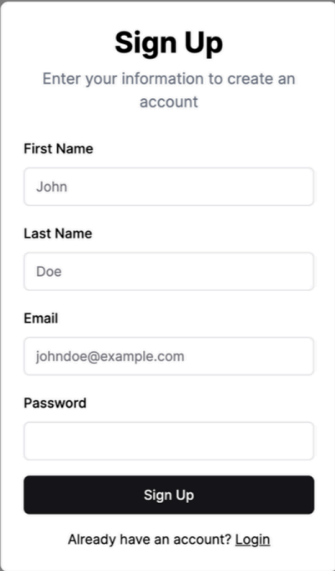
Import `react-router-dom` into your project and add the following routes -

1. `/signup` - The signup page
2. `/signin` - The signin page
3. `/dashboard` - Balances and see other users on the platform.
4. `/send` - Send money to other users

▼ Solution

```
function App() {  
  return (  
    <>  
    <BrowserRouter>  
      <Routes>  
        <Route path="/signup" element={<Signup />} />  
        <Route path="/signin" element={<Signin />} />  
        <Route path="/dashboard" element={<Dashboard />} />  
        <Route path="/send" element={<SendMoney />} />  
      </Routes>  
    </BrowserRouter>  
  )  
}
```

Step 2 - Create and hook up Signup page



Sign Up
Enter your information to create an account

First Name

Last Name

Email

Password

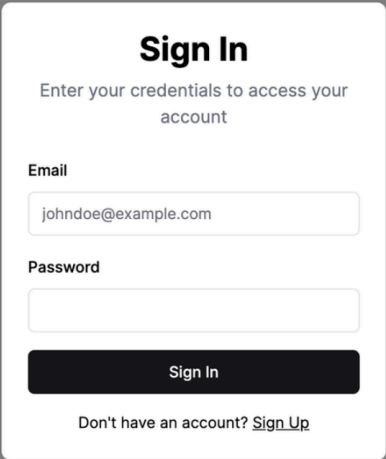
Sign Up

Already have an account? [Login](#)

If the user signup is successful, take the user to `/dashboard`

If not, show them an error message

Step 3 - Create the signin page



Sign In

Enter your credentials to access your account

Email

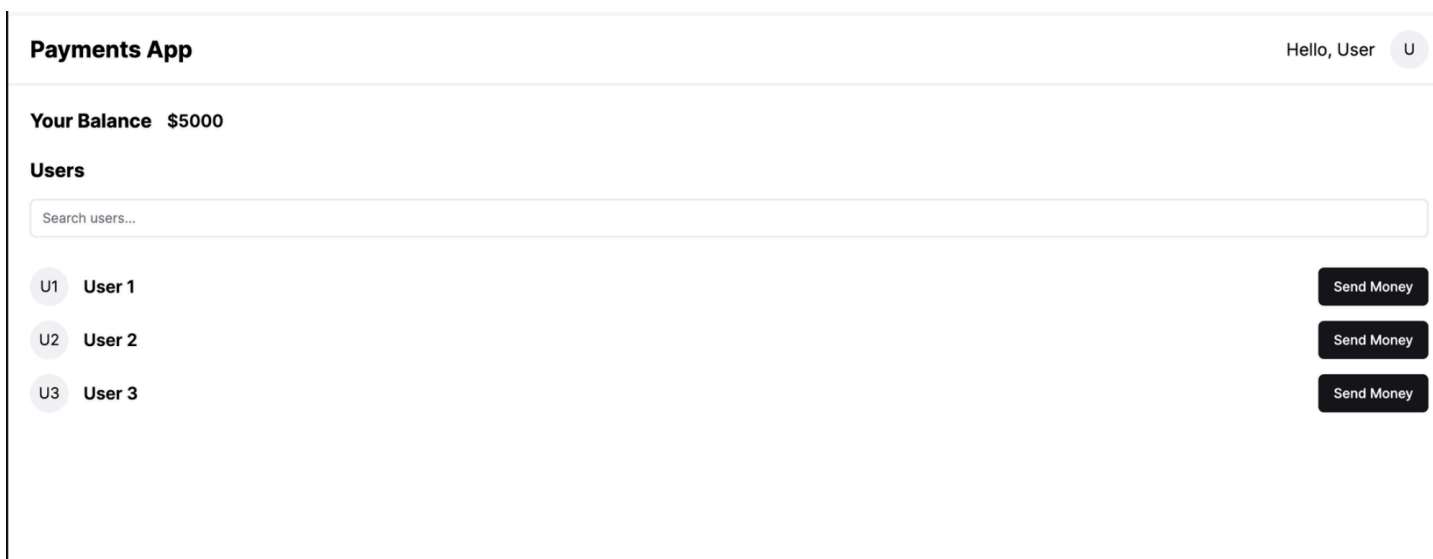
Password

Sign In

Don't have an account? [Sign Up](#)

If the signin is successful, take the user to `/dashboard`

Step 4 - Dashboard page



Payments App Hello, User **U**

Your Balance \$5000

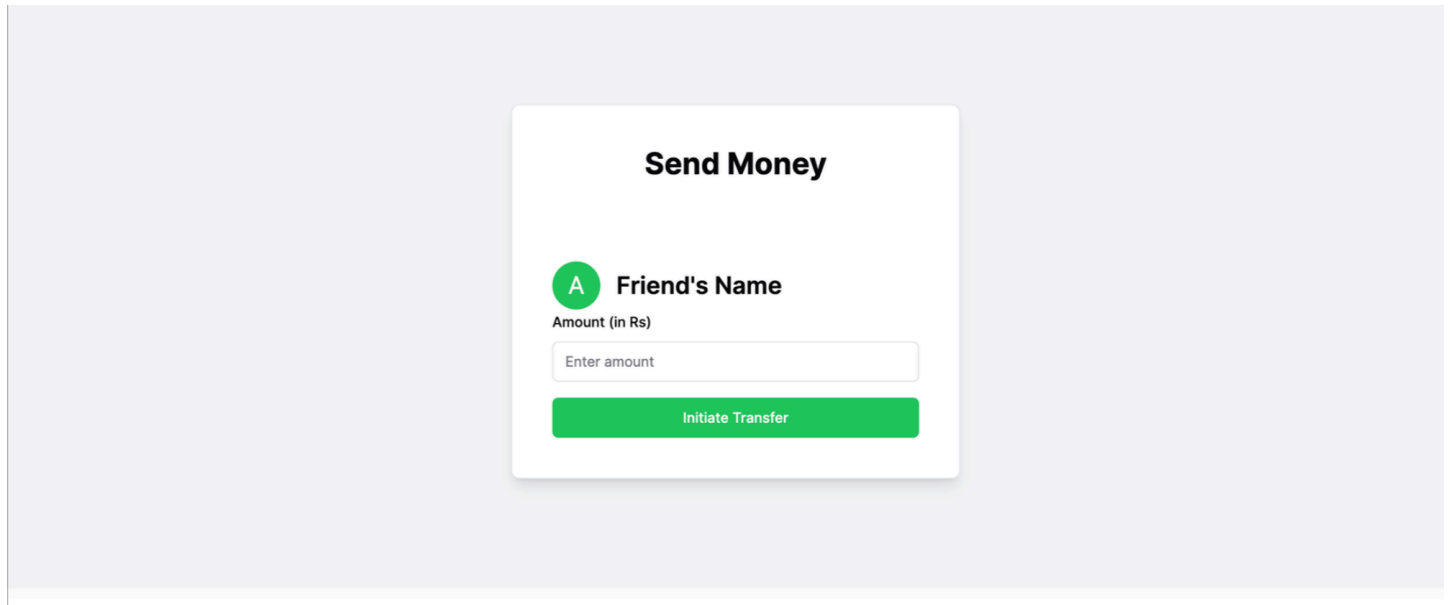
Users

Search users...

U1	User 1	Send Money
U2	User 2	Send Money
U3	User 3	Send Money

Show the user their balance, and a list of users that exist in the database

Clicking on `Send money` should open a modal that lets the user send money



Step 5 - Auth Components

Full Signup component

You can break down the app into a bunch of components. The code only contains the styles of the component, not any onclick functionality.

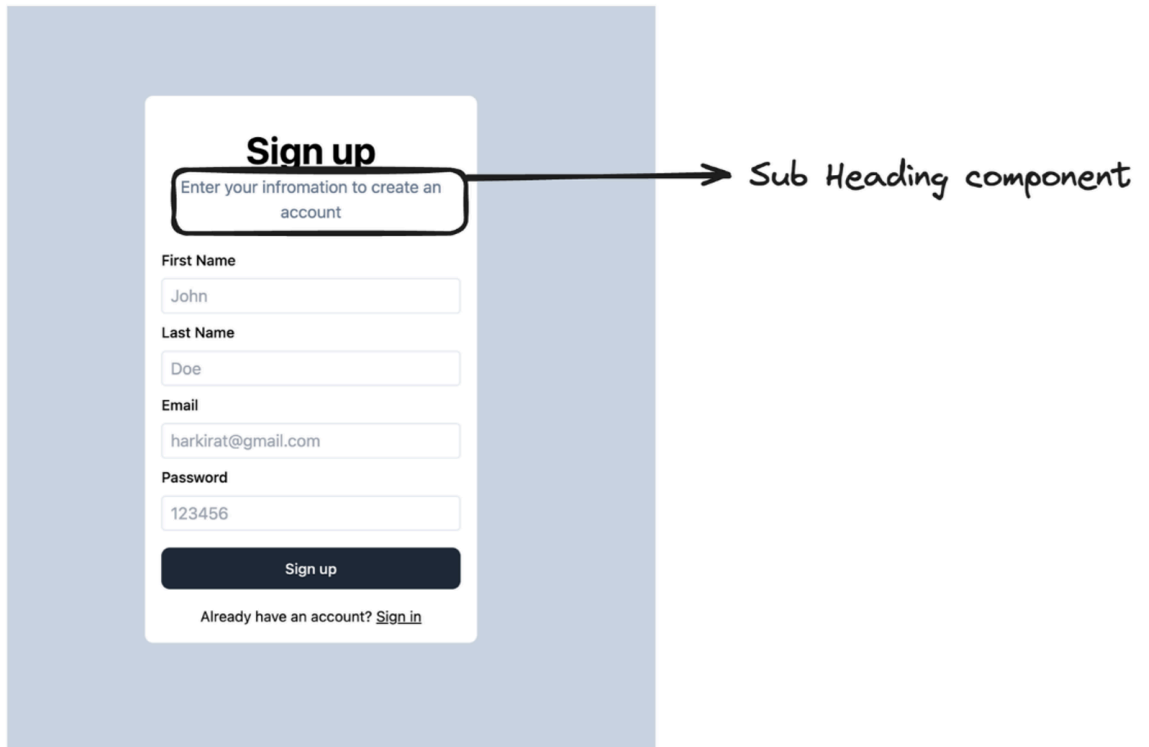
1. Heading component

The image shows a sign-up form on a light blue background. The form is a white card with a rounded rectangle at the top containing the text 'Sign up' in bold. Below this, it says 'Enter your information to create an account'. There are four input fields: 'First Name' (with 'John'), 'Last Name' (with 'Doe'), 'Email' (with 'harkirat@gmail.com'), and 'Password' (with '123456'). At the bottom of the form is a dark blue button with 'Sign up' in white text. Below the button is a link: 'Already have an account? [Sign in](#)'. An arrow points from the 'Sign up' heading to the text 'Heading component' on the right.

▼ Code

```
export function Heading({label}) {  
  return <div className="font-bold text-4xl pt-6">  
    {label}  
  </div>  
}
```

2. Sub Heading component



▼ Code

```
export function SubHeading({label}) {  
  return <div className="text-slate-500 text-md pt-1 px-4 pb-4">  
    {label}  
  </div>  
}
```

3. InputBox component

To move canvas, hold mouse wheel or spacebar while dragging, or use the hand tool

Sign up

Enter your information to create an account

First Name
John

Last Name
Doe

Email
harkirat@gmail.com

Password
123456

Sign up

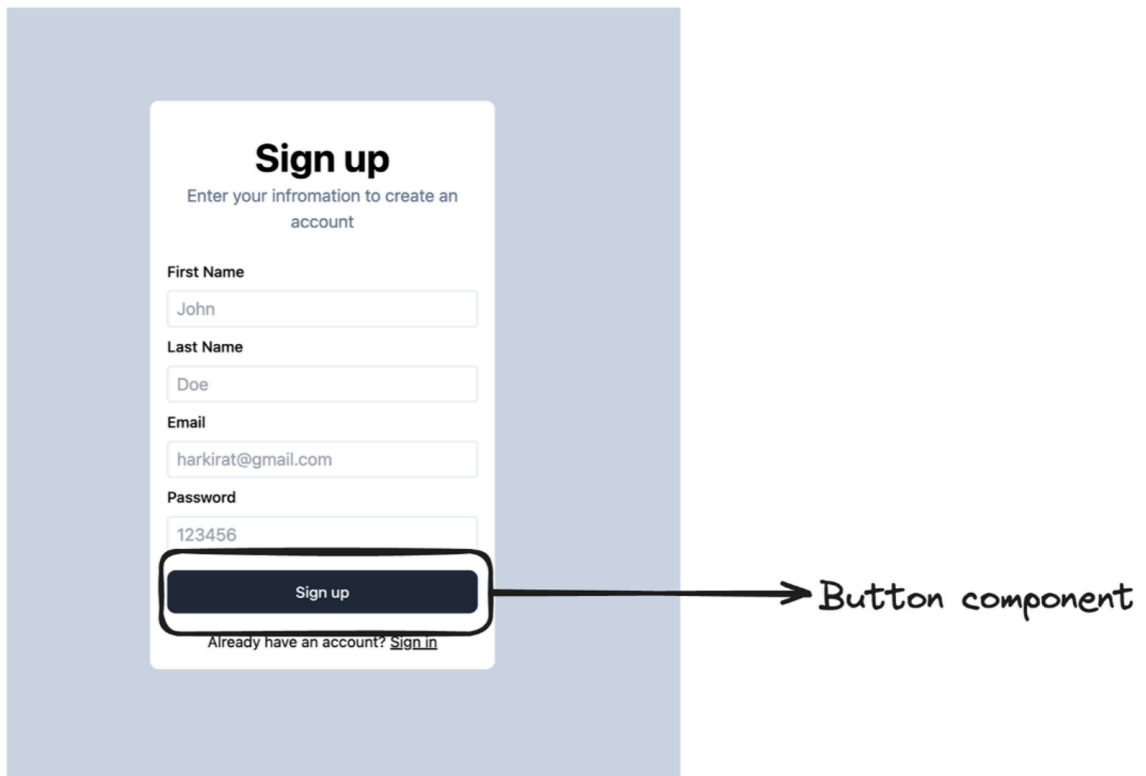
Already have an account? [Sign in](#)

→ InputBox component

▼ Code

```
export function InputBox({label, placeholder}) {  
  return <div>  
    <div className="text-sm font-medium text-left py-2">  
      {label}  
    </div>  
    <input placeholder={placeholder} className="w-full px-2 py-1 border rounded border-slate-300" type="text" />  
  </div>  
}
```

4. Button Component



▼ Code

```
export function Button({label, onClick}) {  
  return <button onClick={onClick} type="button" class=" w-full text-white bg-gray-800 hov  
}
```

This section was blindly copied from <https://flowbite.com/docs/components/buttons/>

5. BottomWarning

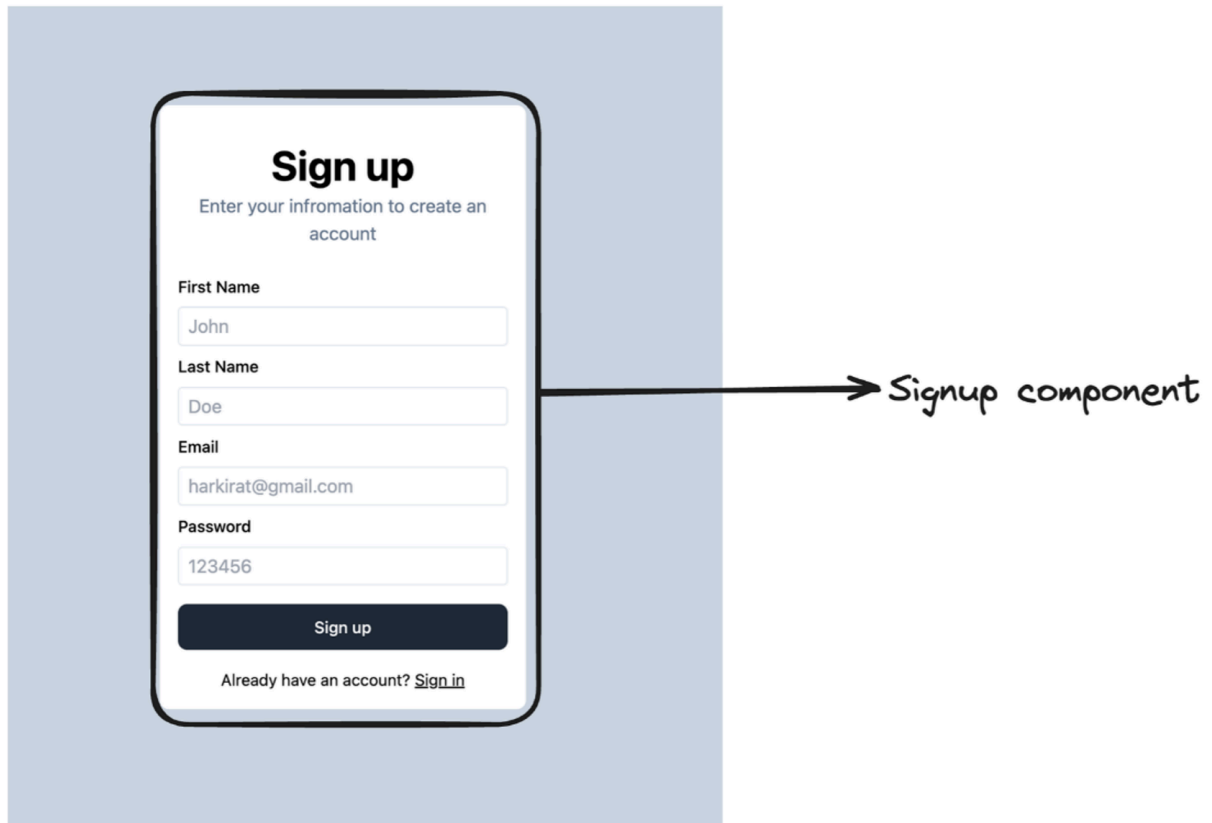


▼ Code

```
import { Link } from "react-router-dom"

export function BottomWarning({label, buttonText, to}) {
  return <div className="py-2 text-sm flex justify-center">
    <div>
      {label}
    </div>
    <Link className="pointer underline pl-1 cursor-pointer" to={to}>
      {buttonText}
    </Link>
  </div>
}
```

Full Signup component



▼ Code

```
import { BottomWarning } from "../components/BottomWarning"
import { Button } from "../components/Button"
import { Heading } from "../components/Heading"
import { InputBox } from "../components/InputBox"
import { SubHeading } from "../components/SubHeading"

export const Signup = () => {
  return <div className="bg-slate-300 h-screen flex justify-center">
    <div className="flex flex-col justify-center">
      <div className="rounded-lg bg-white w-80 text-center p-2 h-max px-4">
        <Heading label={"Sign up"} />
        <SubHeading label={"Enter your information to create an account"} />
        <InputBox placeholder="John" label={"First Name"} />
        <InputBox placeholder="Doe" label={"Last Name"} />
        <InputBox placeholder="harkirat@gmail.com" label={"Email"} />
        <InputBox placeholder="123456" label={"Password"} />
        <div className="pt-4">
          <Button label={"Sign up"} />
        </div>
        <BottomWarning label={"Already have an account?"} buttonText={"Sign in"} to={"/signin"} />
      </div>
    </div>
  </div>
```

```
</div>  
}
```

Full Signin component

▼ Code

```
import { BottomWarning } from "../components/BottomWarning"  
import { Button } from "../components/Button"  
import { Heading } from "../components/Heading"  
import { InputBox } from "../components/InputBox"  
import { SubHeading } from "../components/SubHeading"  
  
export const Signin = () => {  
  
  return <div className="bg-slate-300 h-screen flex justify-center">  
    <div className="flex flex-col justify-center">  
      <div className="rounded-lg bg-white w-80 text-center p-2 h-max px-4">  
        <Heading label={"Sign in"} />  
        <SubHeading label={"Enter your credentials to access your account"} />  
        <InputBox placeholder="harkirat@gmail.com" label={"Email"} />  
        <InputBox placeholder="123456" label={"Password"} />  
        <div className="pt-4">  
          <Button label={"Sign in"} />  
        </div>  
        <BottomWarning label={"Don't have an account?"} buttonText={"Sign up"} to={"/signup"} />  
      </div>  
    </div>  
  </div>  
}
```

Step 6 - Signin-ed Comonents

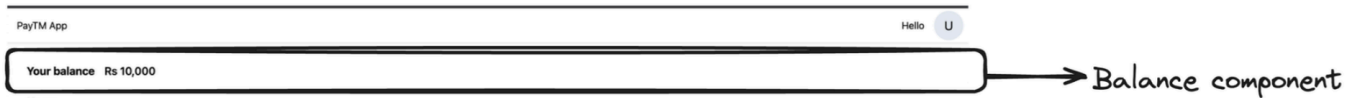
1. Appbar



▼ Code

```
export const AppBar = () => {
  return <div className="shadow h-14 flex justify-between">
    <div className="flex flex-col justify-center h-full ml-4">
      PayTM App
    </div>
    <div className="flex">
      <div className="flex flex-col justify-center h-full mr-4">
        Hello
      </div>
      <div className="rounded-full h-12 w-12 bg-slate-200 flex justify-center mt-1 mr-1">
        <div className="flex flex-col justify-center h-full text-xl">
          U
        </div>
      </div>
    </div>
  </div>
}
```

2. Balance



▼ Code

```
export const Balance = ({ value }) => {  
  return <div className="flex">  
    <div className="font-bold text-lg">  
      Your balance  
    </div>  
    <div className="font-semibold ml-4 text-lg">  
      Rs {value}  
    </div>  
  </div>  
}
```

3. Users component



▼ Code

```
import { useState } from "react"
import { Button } from "../Button"

export const Users = () => {
  // Replace with backend call
  const [users, setUsers] = useState([
    {
      firstName: "Harkirat",
      lastName: "Singh",
      _id: 1
    }
  ])

  return <>
    <div className="font-bold mt-6 text-lg">
      Users
    </div>
    <div className="my-2">
      <input type="text" placeholder="Search users..." className="w-full px-2 py-1 border">
    </div>
    <div>
      {users.map(user => <User user={user} />)}
    </div>
  </>
}

function User({user}) {
  return <div className="flex justify-between">
    <div className="flex">
      <div className="rounded-full h-12 w-12 bg-slate-200 flex justify-center mt-1 mr-1">
        <div className="flex flex-col justify-center h-full text-xl">
          {user.firstName[0]}
        </div>
      </div>
      <div className="flex flex-col justify-center h-full">
        <div>
          {user.firstName} {user.lastName}
        </div>
      </div>
    </div>

    <div className="flex flex-col justify-center h-full">
      <Button label={"Send Money"} />
    </div>
  </div>
}
```

4. SendMoney Component

Send Money

A

Friend's Name

Amount (in Rs)

Initiate Transfer

▼ Code

```
export const SendMoney = () => {
  return <div class="flex justify-center h-screen bg-gray-100">
    <div className="h-full flex flex-col justify-center">
      <div
        class="border h-min text-card-foreground max-w-md p-4 space-y-8 w-96 bg-white
        >
        <div class="flex flex-col space-y-1.5 p-6">
          <h2 class="text-3xl font-bold text-center">Send Money</h2>
        </div>
        <div class="p-6">
          <div class="flex items-center space-x-4">
            <div class="w-12 h-12 rounded-full bg-green-500 flex items-center justify-center">
              <span class="text-2xl text-white">A</span>
            </div>
            <h3 class="text-2xl font-semibold">Friend's Name</h3>
          </div>
          <div class="space-y-4">
            <div class="space-y-2">
              <label
                class="text-sm font-medium leading-none peer-disabled:cursor-not-allowed"
                for="amount"
              >

```

```

        Amount (in Rs)
      </label>
      <input
        type="number"
        class="flex h-10 w-full rounded-md border border-input bg-background
        id="amount"
        placeholder="Enter amount"
      />
    </div>
    <button class="justify-center rounded-md text-sm font-medium ring-offset
      Initiate Transfer
    </button>
  </div>
</div>
</div>
}

```

Step 7 - Wiring up the backend calls

You can use

1. fetch or
2. axios

to wire up calls to the backend server.

The final code looks something like this -

<https://github.com/100xdevs-cohort-2/paytm/tree/complete-solution> (complete-solution branch on the repo)

The important bits here are -

1. Signup call - <https://github.com/100xdevs-cohort-2/paytm/blob/complete-solution/frontend/src/pages/Signup.jsx#L36>
2. Call to get all the users given the filter - <https://github.com/100xdevs-cohort-2/paytm/blob/complete-solution/frontend/src/components/Users.jsx#L13>
3. Call to transfer money b/w accounts - <https://github.com/100xdevs-cohort-2/paytm/blob/complete-solution/frontend/src/pages/SendMoney.jsx#L45>